

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10 – SUB-STRAND 2.3 TRIGONOMETRY: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for our unit. Driving question: How can we use the angles and one side of a right-angled triangle to find any unknown height or distance in our environment, without ever measuring it directly? Answer every section in the space under each prompt. Use what you observed and calculated in our lessons: the school water tank, Amina and Baraka, your paper triangles, your own clinometer and your outdoor measurements. Show your working for every calculation, include units, and write full sentences when you are asked to explain. You may use a calculator. Where a question gives a value from the calculator, you do not need to find it again.

Part 1: The triangle hidden in the water-tank problem

Amina stands 10 m from the base of the school water tank and sights the top of the tank. Draw and label a clear sketch of this situation. Mark the angle of elevation, and label the three sides of the right-angled triangle using the words hypotenuse, opposite and adjacent. Then say which of the three sides you can measure directly with a tape measure from the ground, and which side you cannot measure directly. Explain why 'opposite' and 'adjacent' would swap names if you chose the other acute angle as your reference angle, but 'hypotenuse' would not change.

Write your answer here:

Part 2: Calculating the height from two positions

In Lesson 1, Amina stood 10 m from the base of the tank and measured an angle of elevation of 38° . Baraka stood 20 m from the base and measured an angle of elevation of 22° . Your calculator gives $\tan 38^\circ \approx 0.781$ and $\tan 22^\circ \approx 0.404$. (a) Write down the tangent ratio in words for the angle of elevation in this problem and rearrange it to give a formula for the height h . (b) Use your formula to calculate the height of the tank from Amina's measurements and from Baraka's measurements. Give each answer to 1 decimal place. (c) Compare your two heights. Would you expect them to be exactly equal? Explain your thinking, and say what could cause any difference you find.

Write your answer here:

Part 3: Why does the method work from any distance?

In Lesson 4 you drew three right-angled triangles that each had a 35° angle but had adjacent sides of 5 cm, 8 cm and 11 cm. You found that $\text{opp} \div \text{hyp}$, $\text{adj} \div \text{hyp}$ and $\text{opp} \div \text{adj}$ stayed almost the same in all three triangles, even though the triangles were different sizes. The calculator gave $\sin 35^\circ \approx 0.574$, $\cos 35^\circ \approx 0.819$ and $\tan 35^\circ \approx 0.700$. (a) Match each of the three ratios to its name (sine, cosine, tangent) and write the mnemonic that helps you remember them. (b) Use this finding to explain why the tangent of an angle can be used to find the tank height from any distance. In your answer, say what stays the same for every student and what is different for each student. (c) Your measured ratios were slightly different from the calculator values. Give one reason why.

Write your answer here:

Part 4: The complementary-angle shortcut

The two acute angles in a right-angled triangle add up to 90° . In the water-tank triangle, Amina's angle of elevation is 38° . (a) What is the size of the other acute angle, at the top of the triangle? (b) Using the words opposite and adjacent, explain how the tank height is related to each of these two angles. (c) Use your answer to (b) to explain why $\sin 38^\circ$ and $\cos 52^\circ$ must have the same value. (d) In the table you completed in Lesson 6 you found the difference $\sin \theta - \cos(90^\circ - \theta)$ for each pair of angles. What value did you find in every row, and what does this show?

Write your answer here:

Part 5: Closing the loop: our outdoor measurement and why trigonometry matters

In Lesson 8 you used your own clinometer at two stations, Observer A at 8 m and Observer B at 12 m from the base of the school water-tank tower. (a) Describe the steps you followed to find the height of the tank without climbing it. (b) At which station was your angle of elevation larger, and why? (c) Write a short explanation in CER form (Claim, Evidence, Reasoning) that answers the driving question: how can we find the tank height from a measured distance and an angle? Use your own results from the two stations as your evidence. (d) The KICD key inquiry question asks: Why is trigonometry important? Give your answer using two examples of people who use this method in Kenya.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Naming the sides and using SOH-CAH-TOA	Labels opposite, adjacent and hypotenuse correctly from the chosen angle, explains why the names swap, and states sine, cosine and tangent with the SOH-CAH-TOA definitions accurately.	Labels the sides correctly for the water-tank triangle and states the three ratios, but the explanation of the swapping of names is incomplete.	Mislabeled sides or confuses the ratios, and gives little or no explanation of the reference angle.
Calculation with $h = d \times \tan \theta$	Rearranges the tangent ratio correctly, calculates both heights accurately (about 7.8 m and 8.1 m) with units, and shows all working.	Uses the formula correctly with only a minor slip in rounding, units or working.	Does not rearrange the formula correctly or makes calculation errors that change the result.
Explaining consistency and the constant ratio	Clearly explains that the ratio depends only on the angle, that different distances give different angles and tan values but the same height through $h = d \times \tan \theta$, and attributes small differences to measurement uncertainty.	States that the ratio is constant and the heights are similar, but the reasoning about why is partly stated.	Says only that the answers are 'the same' or 'different' with no mathematical reasoning.
Complementary angles and CER explanation with Kenyan contexts	Explains $\sin \theta = \cos(90^\circ - \theta)$ using opposite and adjacent, and writes a full CER using the 8 m and 12 m results, with	Gives the complementary-angle result and a CER with claim and evidence, but the reasoning or the examples are thin.	States the result without explanation, and the CER is missing parts or does not use the outdoor evidence.

	two accurate Kenyan examples of use.		
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